

At the venue, before flight

SORTIE # _____

ArduPilot Copter · Mission Planner · **Stabilize** · everything else was done on the bench, see Checklist 1

01 What the walk may have broken

Transit only

- ☐ **Shake test** — nothing rattles, nothing swings, nothing clicks
- ☐ Every connector still seated: ESCs, PM02, **GPS1**, **TELEM1**, **GPS2**

02 Fit the propellers

Battery disconnected

- ☐ **Look at the battery lead and confirm it is disconnected.** Do not rely on remembering
- ☐ CW propellers on the CW motors. **Diagonally opposite motors turn the same way** — front-right and rear-left counter-clockwise, front-left and rear-right clockwise. Confirm against the frame diagram in Mission Planner

03 Power up

The order matters

- ☐ Aircraft on the floor mid-box, **nose pointing away from people**
- ☐ Battery fitted and strapped, **then** connected — announce **"POWERING UP"**, and step out of the propeller arc

04 Pre-arm

On the ground station

- ☐ **No **PreArm:** message** in Mission Planner. If there is one, read it. Do not click past it
- ☐ Flight mode reads **Stabilize** on the switch position the pilot will use
- ☐ **Attitude check:** tilt the airframe nose-up — the horizon moves the right way; roll it left — it rolls left
- ☐ RC link healthy — move each stick and watch it move in Mission Planner

05 Numbers to monitor

Mission Planner, during every sortie

Read aloud while the aircraft is up. **Any one of them out of range ends the sortie** — land first, work out why afterwards.

- ☐ **Battery voltage** — land at **14.4 V** under load. Below 13.6 V you are already late
- ☐ **Vibration** VibeX/Y/Z — **under 30 m/s²**. 30–60 is marginal, over 60 will not fly properly
- ☐ **Clipping** Clip0/1/2 — **must stay 0**. Any increase means the airframe is shaking the IMU past its range
- ☐ **Altitude** — **trend only**. Barometer alone, so ±1 m of indoor drift is normal
- ☐ **EKF variance** — **all bars under 0.5**, green. Amber or red means the estimator does not trust itself
- ☐ **Motor outputs** RCOU 1–4 — **within ~10%** in a steady hover. A persistent split across a diagonal means CG or a twisted arm